

Operating Systems

CSIT — 4th Semester — 2082 — Answer Sheet
Subject Code: CSC264

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Explain the differences between deadlock, livelock and starvation. A system has four processes and five allocated resources. Given the following current allocation and maximum needs, find the smallest value of x for which this is a safe state.

Process	Allocated	Maximum	Available
A	10	21	11
B	11	12	21
C	10	22	10
D	11	10	21
E	11	11	22

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How the files are implemented using linked list allocation? How many disk operations are needed to fetch the i-node for a file with the path name /usr/ast/courses/os/handout.t? Assume that the i-node for the root directory is in memory, but nothing else along the path is in memory. Also assume that all directories fit in one disk block.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. If a disk has double interleaving, does it also need cylinder skew in order to avoid missing data when making a track-to-track seek? Justify your answer. Consider a disk queue with requests for I/O to blocks on cylinders 24, 90, 133, 43, 188. The head is initially at cylinder number 100 and the cylinders are numbered from 0 to 199. Calculate the total head movement for servicing the request using FCFS, SSTF and CSCAN.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. List and describe two main functions of operating system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define segmentation. Why do we need multiple queues in process scheduling?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What do you mean by open source operating systems? Why do you think the designers of Linux made it impossible for a process to send a signal to another process that is not in its process group?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. During implementing mutual exclusion, how can you solve the problem of busy wait? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is race condition? Explain how time quantum value and context switching time affect each other, in a round-robin scheduling algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Five batch jobs A, B, C, D and E, with estimated running times 10, 6, 2, 4 and 8 minutes respectively, arrived at the same time. Determine the average turnaround time for the scheduling algorithms FCFS and SJF.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Given the page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 7, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6, compute the numbers of page faults for FIFO page replacement algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is Belady's anomaly? For the decimal virtual addresses, 20000, 32168, 60000, compute the virtual page number and offset for a 4-KB page and for an 8 KB page.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Differentiate between a) Preemptive and non-preemptive scheduling. b) Physical address and logical address.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.